

The Origins of Time-Series Timeability:

Cross-Asset AR–GARCH Evidence

—Why CSI 1000 Timing Works, and Where It Can Be Replicated

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Abstract

Using daily data on 23 assets (Chinese broad equity indices, precious metals, commodity futures, overseas indices, and individual A-share stocks), we estimate an AR(1)–GARCH(1,1) model augmented with state-interaction terms to identify the statistical origins of time-series timing ability. Main findings: (i) the short-horizon momentum coefficient φ_1 has a rank correlation of 0.77 ($p < 0.01$) with realized timing-strategy performance, making it the primary discriminating variable; (ii) the conditional-volatility level term φ_2 (the “volatility premium”) is about +11bp/day per standard deviation for the CSI 1000—the strongest in the sample—mirroring the fact that roughly 60% of the genetically mined factor library consists of volatility-family factors; (iii) volatility predictability (GARCH persistence, Mincer–Zarnowitz R^2) is *unrelated* to timing performance; (iv) at the individual-stock level φ_1 is uniformly negative: index-level positive autocorrelation is an emergent property of cross-sectional aggregation (lead–lag effects); (v) out-of-sample validation: the SZSE 2000 and CSI 2000 exhibit parameter structures nearly identical to—and stronger than—the CSI 1000, making them the top candidates for extending the framework.

1 Model

The mean and volatility equations for log returns r_t (in %) are

$$r_t = \mu + \varphi_1 r_{t-1} + \varphi_L \tilde{m}_{t-1} + \varphi_2 \tilde{\sigma}_{t-1} + \varphi_3 (\tilde{\sigma}_{t-1} \cdot r_{t-1}) + \varepsilon_t, \quad (1)$$

$$\varepsilon_t = \sigma_t e_t, \quad e_t \sim \text{i.i.d. } N(0, 1), \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (2)$$

where $\tilde{\sigma}_{t-1}$ is the standardized conditional volatility (the state variable), $\tilde{m}_{t-1}$ is the standardized cumulative return over the past 244 trading days (long-horizon momentum), and the interaction $\tilde{\sigma}_{t-1} \cdot r_{t-1}$ captures volatility-state modulation of short-horizon autocorrelation: $\varphi_1 + \varphi_3 \tilde{\sigma}_{t-1}$ is the time-varying first-order autocorrelation.

Estimation is in two steps: first, an AR(1)–GARCH(1,1) extracts the conditional volatility $\hat{\sigma}_t$ used to construct the state variables; second, these enter the mean equation as exogenous regressors together with the interaction term, with GARCH(1,1) disturbances, estimated by QMLE. We report Bollerslev–Wooldridge robust t -statistics. All series start in 2010; observations with $|r_t| > 25\%$ are discarded.

Interpretation. $\varphi_1 > 0$: daily momentum (yesterday’s gain predicts today’s gain); $\varphi_2 > 0$: positive risk compensation in high-volatility states (“panic-then-rebound”); $\varphi_L > 0$: continuation of annual momentum; $\varphi_3 < 0$: stronger reversal in high-volatility states.

2 Data

Chinese index data, commodity front-month continuous contracts, and spot gold (au9999) come from the project’s own price library; the SZSE 2000, CSI 2000, BSE 50, and six individual A-shares (split-adjusted) come from the EastMoney history API. The stocks span the size/investor-structure spectrum: Kweichow Moutai (mega-cap), Ping An Insurance (large-cap), Hualu Hengsheng (mid-cap), Innolight (large ChiNext), Leo Group and Guanhua Sci-Tech (small-caps).

3 Estimation Results

Table 1: Chinese broad equity indices: five-term mean equation with GARCH(1,1) errors

	CSI 300	CSI 500	CSI 1000	ChiNext	SZSE 2000	CSI 2000	BSE 50
$\hat{\mu}$ const. (ann. %)	3.0 (0.55)	6.5 (1.03)	10.1 (1.46)	13.9* (1.81)	13.0* (1.92)	11.3 (1.42)	−4.3 (−0.17)
$\hat{\varphi}_1$ (r_{t-1})	0.0151 (0.84)	0.0376* (1.95)	0.0550*** (2.70)	0.0312* (1.72)	0.0538*** (2.61)	0.0540** (2.32)	−0.0051 (−0.10)
$\hat{\varphi}_L$ ($\tilde{m}_{t-1}$, bp)	1.2 (0.44)	5.2 (1.49)	5.4 (1.47)	4.9 (1.08)	7.5** (2.11)	10.6** (2.53)	23.8 (0.81)
$\hat{\varphi}_2$ ($\tilde{\sigma}_{t-1}$, bp)	−1.4 (−0.52)	6.0* (1.81)	11.0*** (2.84)	7.9** (2.03)	10.8*** (2.92)	10.8** (2.46)	10.8 (0.83)
$\hat{\varphi}_3$ ($\tilde{\sigma}_{t-1} \cdot r_{t-1}$)	−0.0206 (−1.17)	0.0032 (0.17)	0.0051 (0.25)	−0.0035 (−0.21)	0.0019 (0.09)	0.0009 (0.04)	0.0092 (0.29)
$\hat{\alpha} + \hat{\beta}$	0.981	0.952	0.933	0.978	0.929	0.909	0.975
MZ R^2	0.115	0.138	0.135	0.106	0.114	0.112	0.110
VR(5)	1.022	1.100	1.147	1.059	1.156	1.205	1.113
B&H Sharpe	0.07	0.13	0.20	0.26	0.20	0.30	0.05
Ann. vol (%)	21	25	27	30	26	28	38
N	3771	3771	3527	3671	3530	2806	853
Sample start	2010-01	2010-01	2010-01	2010-06	2011-01	2014-01	2022-05

Bollerslev–Wooldridge robust t -statistics in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels (two-sided).

$\hat{\varphi}_L, \hat{\varphi}_2$ are in bp/day per one-s.d. move of the state variable; the long-momentum window is $\min(244, N/6)$ days.

Tables 1–3 report the group estimates; Table 4 correlates each parameter with the realized performance of the project’s live timing strategy (genetically mined factor library with IC weighting, backtested 2018–2026). Four conclusions:

(1) Short-horizon momentum φ_1 is the primary discriminator. Rank correlation +0.77 ($p = 0.004$). Chinese broad indices display a monotone gradient—CSI 1000 (+0.055, $t = 2.7$) > CSI 500 > CSI 300 (insignificant)—exactly matching the realized ordering “explosive > moderate > weak”. Commodities have uniformly negative φ_1 (daily reversal of 4–6bp, below round-trip costs); US indices are negative with high drift.

(2) The volatility premium φ_2 is the second source. CSI 1000: +11.0bp/day ($t = 2.8$); CSI 300: negative. This maps directly onto the factor library’s composition: about 60% of

Table 2: Precious metals and commodity futures (front-month continuous)

	Gold spot	Gold fut.	Silver	Rebar	Copper	Soy meal	PTA	Methanol
$\hat{\mu}$ const. (ann. %)	15.0*** (3.27)	6.7* (1.70)	6.6 (0.90)	-1.5 (-0.26)	4.7 (1.04)	-2.9 (-0.45)	-3.2 (-0.54)	4.7 (0.55)
$\hat{\varphi}_1$ (r_{t-1})	0.0156 (0.62)	-0.0079 (-0.42)	0.0058 (0.29)	-0.0440** (-2.32)	-0.0438** (-2.53)	-0.0530** (-2.12)	0.0130 (0.68)	-0.0436** (-2.10)
$\hat{\varphi}_L$ ($\tilde{m}_{t-1}$, bp)	0.1 (0.04)	1.5 (0.82)	-1.5 (-0.39)	1.1 (0.34)	0.5 (0.26)	-5.4 (-1.57)	-0.8 (-0.32)	-0.1 (-0.03)
$\hat{\varphi}_2$ ($\tilde{\sigma}_{t-1}$, bp)	2.0 (0.46)	-0.5 (-0.28)	2.4 (0.56)	2.2 (0.82)	1.2 (0.55)	-0.9 (-0.30)	2.4 (1.02)	-2.7 (-0.69)
$\hat{\varphi}_3$ ($\tilde{\sigma}_{t-1} \cdot r_{t-1}$)	-0.0216 (-0.87)	-0.0188 (-0.97)	-0.0111 (-0.61)	-0.0251 (-1.62)	-0.0473*** (-3.03)	0.0236 (1.31)	-0.0659*** (-3.97)	-0.0253 (-1.45)
$\hat{\alpha} + \hat{\beta}$	1.046	1.006	1.007	0.987	0.951	1.015	1.001	1.006
MZ R^2	0.082	0.072	0.185	0.087	0.104	0.009	0.079	0.100
VR(5)	0.930	0.922	1.058	0.935	0.952	0.882	0.975	0.916
B&H Sharpe	0.84	0.48	0.21	-0.10	0.18	0.00	-0.09	0.11
Ann. vol (%)	15	16	28	23	19	22	24	29
N	2077	3776	3206	3767	3773	3771	3764	2556
Sample start	2016-12	2010-01	2012-05	2010-01	2010-01	2010-01	2010-01	2014-12

Bollerslev–Wooldridge robust t -statistics in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels (two-sided).

$\hat{\varphi}_L, \hat{\varphi}_2$ are in bp/day per one-s.d. move of the state variable; the long-momentum window is $\min(244, N/6)$ days.

Table 3: Overseas indices and individual A-share stocks (adjusted prices)

	Nasdaq 100	S&P 500	Moutai	Ping An	Hualu	Leo Grp	Innolight	Guanghua
$\hat{\mu}$ const. (ann. %)	21.8*** (3.65)	18.7*** (4.72)	39.6** (2.20)	20.8 (0.97)	74.0*** (2.89)	-4.9 (-0.37)	67.8*** (2.65)	-4.4 (-0.28)
$\hat{\varphi}_1$ (r_{t-1})	-0.0309 (-1.39)	-0.0403** (-1.96)	-0.0193 (-0.79)	-0.0236 (-1.04)	-0.0552*** (-2.73)	0.0206 (1.02)	0.0629*** (2.88)	0.0097 (0.42)
$\hat{\varphi}_L$ ($\tilde{m}_{t-1}$, bp)	-1.0 (-0.39)	-1.9 (-1.39)	-2.5 (-0.26)	-4.0 (-0.57)	-11.8 (-0.69)	6.4 (0.97)	6.2 (0.73)	-2.9 (-0.45)
$\hat{\varphi}_2$ ($\tilde{\sigma}_{t-1}$, bp)	5.4 (1.62)	6.7** (2.53)	25.7* (1.73)	10.8 (0.67)	44.8** (1.97)	1.5 (0.19)	20.0 (1.18)	-6.3 (-0.69)
$\hat{\varphi}_3$ ($\tilde{\sigma}_{t-1} \cdot r_{t-1}$)	-0.0605*** (-3.90)	-0.0322** (-2.22)	0.0132 (0.32)	-0.0257 (-0.69)	-0.0409* (-1.86)	0.0469** (2.47)	0.0318 (1.16)	0.0075 (0.29)
$\hat{\alpha} + \hat{\beta}$	0.890	0.836	0.985	0.987	1.001	0.975	0.993	0.923
MZ R^2	0.267	0.352	0.100	0.186	0.299	0.240	0.203	0.154
VR(5)	0.826	0.857	0.947	0.993	0.837	1.204	1.081	1.071
B&H Sharpe	0.76	0.66	0.45	0.30	0.59	0.21	0.77	0.25
Ann. vol (%)	21	17	53	69	86	53	83	52
N	2875	3917	2161	2817	2775	3435	2726	2485
Sample start	2014-02	2010-01	2016-07	2010-01	2010-01	2010-01	2012-04	2015-02

Bollerslev–Wooldridge robust t -statistics in parentheses; ***, **, * denote significance at the 1%, 5%, 10% levels (two-sided).

$\hat{\varphi}_L, \hat{\varphi}_2$ are in bp/day per one-s.d. move of the state variable; the long-momentum window is $\min(244, N/6)$ days.

Table 4: Spearman rank correlation between model parameters and realized strategy performance ($n = 12$)

Parameter	$\hat{\rho}$	p -value
$\hat{\varphi}_1$ (r_{t-1} , short momentum)	+0.766***	0.004
$\hat{\varphi}_L$ ($\tilde{m}_{t-1}$, long momentum)	+0.689**	0.013
$\hat{\varphi}_2$ ($\tilde{\sigma}_{t-1}$, volatility premium)	+0.461	0.132
$\hat{\varphi}_3$ ($\tilde{\sigma}_{t-1} \cdot r_{t-1}$, interaction)	+0.377	0.227
$\hat{\alpha} + \hat{\beta}$ (GARCH persistence)	−0.434	0.158
MZ R^2 (vol. predictability)	+0.362	0.248
VR(5) (variance ratio)	+0.347	0.270
B&H Sharpe	+0.488	0.108

Realized-performance ranks: CSI 1000 = 4, CSI 500 = 3, weak/mediocre = 2, commodities = 1; assets without live results are excluded.

in-service zz1000 factors are volatility-family (signals highly correlated with rolling volatility, direction “long when volatility is high”), and all top-20 factors by cos IC belong to this family. The market’s structure is what the genetic search grows into.

(3) Volatility predictability is unrelated to timeability. Rank correlations of GARCH persistence and MZ R^2 are insignificant. Moutai and the Nasdaq 100 have the most predictable volatility (MZ $R^2 \approx 0.27$ – 0.32) yet the least predictable direction. Being able to forecast volatility is not the same as being able to time.

(4) Stock-level reversal coexists with index-level momentum: an emergent property of aggregation. All six stocks have negative φ_1 (Hualu Hengsheng: -0.055 , $t = -2.7$ —same magnitude as the CSI 1000 with the opposite sign). Index-level positive autocorrelation does not arise from momentum in the constituents; idiosyncratic reversals cancel in the cross-section, leaving the lead–lag propagation of the common component (Lo–MacKinlay). *Timeability lives at the aggregate level and does not survive disaggregation.*

4 Mechanisms and Corroborating In-Project Evidence

Factor-library taxonomy. Classifying each of the 79 in-service zz1000 factors by the regression feature its signal correlates with most: 47 volatility-family (mean cos IC +0.079), 17 momentum-family (mean -0.018), 0 interaction-family. The absence of interaction factors is consistent with the insignificant φ_3 ($t = 0.25$): the structure does not exist in the market, and evolution correctly failed to produce it—evidence of *not* overfitting.

Long momentum and the direction lock-in of the “veteran” factors. φ_L is weakly positive for the CSI 1000 (+5.4bp, $t = 1.5$). The project’s nine hand-crafted momentum factors (“veterans”) had their signs fixed at -1 once and for all during the 2013–14 warm-up and were never re-evaluated—precisely opposite to the post-2015 positive long momentum. Under the identical transformation, the raw momentum signal has cos IC +0.051 while the veterans realize -0.054 : mirror images. Removing the seven in-service veterans raises portfolio annualized return from 20.7% to 22.8% (Sharpe 1.01 $\rightarrow$ 1.11); switching to cos IC weighting with veterans excluded yields 24.9% (Sharpe 1.22).

5 An Operational Definition of Timeability, and Extensions

Definition (for daily momentum-family long-short timing algorithms): an asset is timeable if and only if (T1) $\varphi_1 > 0$ with $t > 2$ and $\text{VR}(5) > 1.05$; (T2) $\varphi_2 > 0$ with $t > 1.5$; (T3) buy-and-hold Sharpe $|SR| < 0.3$ (low drift, so timing pays no “drift tax”).

Of the 23 assets, only the CSI 1000, CSI 500 (each item one notch weaker), and the **SZSE 2000** and **CSI 2000** satisfy all three. The CSI 2000 is at least as strong as the CSI 1000 on every item ($\varphi_1 = +0.054$, $\varphi_2 = +10.8\text{bp}$, $\varphi_L = +10.6\text{bp}$ and significant, $\text{VR}(5) = 1.205$ —the sample maximum), making it the top extension candidate. The BSE 50 sample is too short (853 days) to evaluate. Caveat: the 2000-tier indices have no index futures; the short leg must be implemented long-only or proxy-hedged with IM (CSI 1000) futures.

6 Limitations

(i) The state variables are generated regressors, so the t -statistics on φ_2, φ_3 are mildly overstated (Pagan, 1984); (ii) the rank-correlation sample contains only 12 assets, with ordinal, subjectively assigned performance ranks; (iii) gold’s timing performance is concentrated after 2022, consistent with our finding that all four of its predictive terms are insignificant while drift is high—those gains should be attributed to trend exposure, not timing skill; (iv) all regressions are full-sample; conclusions for the extension candidates (2000-tier) await rolling out-of-sample backtests.